

15/12/23

Physics

Ans 1. (3) concave mirror

Ans 2. $u = -10 \text{ cm}$

$$m = -3$$

$$m = \frac{v}{u}$$

$$-3 = \frac{v}{-10}$$

$$v = 30 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{30} - \left(\frac{1}{-10} \right) = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1+3}{30}$$

$$\frac{1}{f} = \frac{4}{30}$$

$$f = \frac{30}{4}$$

$$f = 7.5 \text{ cm}$$

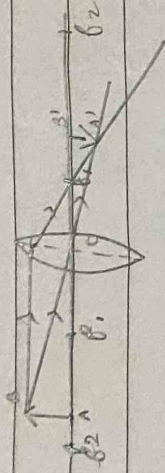
(3) 30 cm, 7.5 cm

Ans 3. (4) Both 2 and 3

Ans 4. (i) Focus :- After reflection where the points converge in a concave mirror and appear to diverge from in a convex mirror to form an image is called the focus or principal focus as it lies on the principal axis, It is denoted by 'F'.

(ii) Centre of Curvature :- The centre of curvature is the centre of the sphere of which the spherical mirror is a part. It is denoted by 'C'.

Ans 5:



$$v = +18$$

$$m = \frac{h_i}{h_o} = \frac{v}{u} = -\frac{5}{18}$$

$$\frac{18}{u} = -\frac{5}{18}$$

$$18 \times 18 = -u$$

$$u = -90$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{18} + \frac{1}{90}$$

$$\frac{1}{f} = \frac{5+1}{90}$$

$$f = \frac{90}{6}$$

$$f = 15 \text{ cm}$$

The nature of the lens is a convex lens, Converging lens.

Ans 6: (i) $h_o = 7 \text{ cm}$

$$u = -27 \text{ cm}$$

$$f = -18 \text{ cm}$$

By Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-18} - \frac{1}{-27}$$

$$\frac{1}{v} = \frac{-3+2}{54}$$

$$\frac{1}{V} = -\frac{1}{54}$$

$$V = -54 \text{ cm}$$

$$m = \frac{h_i}{h_o} = -\frac{V}{u}$$

$$\frac{h_i}{7} = -\frac{(-54)}{-27}$$

$$h_i = \frac{2 \cdot 54 \times 7}{-27},$$

$$h_i = -14$$

$$m = -\frac{14}{7}$$

$$m = -2$$

∴ The image is magnified and its a real image.
and the screen should be placed 54 cm
in front of the mirror to get the image.

$$(ii) R = 5 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$f = \frac{R}{2} = \frac{5}{2} = 2.5 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} - \frac{1}{u} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{2.5} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{10}{25} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{40+5}{100}$$

$$v = \frac{100}{45}$$

$$v = 2.22 \text{ cm}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$= \frac{+2.22}{+20}$$

$$= 0.11$$

∴ The image is shorter than the object, the image will be smaller, and the nature of image is virtual, upright and diminished. //

(iii) The mirror used by the dentist is concave mirror. This is because concave mirror forms a virtual, erect and enlarged image when the object is placed within focus.

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